

ADITYA GULATI

☎ (617) 708-9276 ✉ gulati.adi@northeastern.edu linkedin.com/in/aditya-gulati-2601 github.com/Adityagulatii

EDUCATION

Northeastern University <i>Master's in Artificial Intelligence</i>	Boston, MA <i>Expected May 2028</i>
University of Toronto <i>Honors Bachelor of Science in Statistics and Mathematics</i>	Toronto, ON <i>June 2022</i>

RELEVANT EXPERIENCE

NU Launch Lab – WayfinderAI <i>Co-Lead, Computer Vision Engineer</i>	Jan. 2026 – Present <i>Boston, MA</i>
– Led a 3-person team to deliver a working AI indoor navigation system from scratch, shipping a live Vercel demo within 8 weeks; won 1st Place at Hack@URI AI Track out of 30+ teams. – Fine-tuned Qwen3-VL-8B via 3-stage LoRA + GRPO for floorplan-to-navigation-graph generation; deployed YOLOv8 shelf detection (mAP@50 of 0.893) and EasyOCR aisle sign recognition. – Built graph-based pathfinding in Python using NetworkX and Dijkstra's algorithm to route users across multi-floor buildings without GPS, delivered as a React PWA with real-time map overlays.	
Northeastern Satellite Lab – CubeSat Star Tracker <i>Software Engineer – Computer Vision</i>	Jan. 2026 – Present <i>Boston, MA</i>
– Designed a centroid-based attitude determination pipeline in Python for Raspberry Pi with INT8 quantization, reducing model size by 75% while meeting strict satellite power constraints. – Built a synthetic sky simulator using the 118,000+ star Hipparcos catalog to generate training data across varied sensor noise and orbital conditions, validated across 1,899 ISS-orbit passes.	
Climate Resilient Communities <i>GIS Analyst</i>	Apr. 2024 – Sept. 2025 <i>Toronto, ON</i>
– Built an ArcGIS Pro toolkit using SVM-based land classification and regression-based property valuation across 5+ urban districts, cutting manual climate risk assessment time by an estimated 40%. – Applied spatiotemporal clustering to vegetation, heat index, and emergency route data across 3 regions, producing prioritized risk maps adopted by city planning teams.	
Openlane <i>Data Analyst</i>	Oct. 2022 – Mar. 2024 <i>Toronto, ON</i>
– Engineered ML-based data cleaning pipelines in Python using regression and KNN across 100,000+ records, raising dataset quality from 80% to 95% and reducing downstream reporting errors. – Managed CRM workflows and data validation across 10+ major client accounts, directly contributing to 30% business growth through improved data reliability.	
University of Toronto <i>Teaching Assistant / Learning Objects Assistant</i>	Sept. 2021 – Aug. 2022 <i>Toronto, ON</i>
– Co-taught Data Visualization (Excel, Power BI) and built an R grading-automation script (98% accuracy, 70% faster). – Produced 20 Excel/VBA video tutorials for a management training program.	
BEAR – Behavioural Economics in Action at Rotman <i>Quantitative Research Assistant</i>	May 2021 – Sept. 2021 <i>Toronto, ON</i>
– Ran statistical analysis in Excel on behavioral experiment data, identifying and resolving bugs that improved dataset accuracy by 70%. – Informed the reevaluation of research hypotheses based on the corrected dataset findings.	
University of Toronto Varsity Blues <i>Data Analyst</i>	Oct. 2020 – May 2021 <i>Toronto, ON</i>
– Conducted traffic analysis with Google Analytics to optimize marketing strategy and ad timing. – Increased website traffic and event registrations by up to 20% for varsity campaigns.	

PROJECTS

PitLane AI – Motorsport Coaching Platform

Jan. 2026 – Present

Founder, ML Engineer

Boston, MA

- Built encoder-decoder video pipeline combining GoPro/GPMF telemetry with VideoLLaMA 3 scene analysis and Groq LLaMA coaching text generation, delivering personalized lap feedback for karting drivers, bikers, and riders.
- Won **Voxel51** and **Twelves Computer Vision Hackathon** at Northeastern, and **Cursor Boston Hackathon** for the project.

SKILLS

Languages: Python, R, SQL, C++, JavaScript

Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV

Models & Techniques: VLMs, CNNs, LLMs, Object Detection, RAG, SVM, KNN, Regression, Spatiotemporal Clustering

Tools: NumPy, Pandas, LangChain, CUDA, Git, Docker, Linux, React, FastAPI, Ollama

Cert: ArcGIS Pro – Esri (2024)